higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CONTROL
NQF LEVEL 3

13 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 True
1.2 True
1.3 True
1.4 True
1.5 True
1.6 True
1.7 True
1.8 False
1.9 False
1.10 False

(10 × 1) **[10]**

QUESTION 2

2.1 alarm
2.2 controller
2.3 primary device
2.4 recording
2.5 error

(5 × 1) **[5]**

QUESTION 3

3.1 B
3.2 F
3.3 H
3.4 E
3.5 D

(5 × 1) **[5]**

QUESTION 4

4.1 To show whether or not a process is running within its limits (2)

4.2 The upper control limit is the top limit in the control of data points that fall above the average line in a control chart. (2)

4.3 The operating range is between 21 kPa and 103 kPa or between 20 KPa and 100 KPa. (2)

4.4 To make sure that for a reading of zero, the indicator shows a value of zero, i.e. adjusting the indicator to zero before taking measurements (2)

4.5 To find the difference between the actual reading and the standard, i.e. checking how accurate the instrument is (2)

[10]

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QUESTION 5

5.1 A sight glass operates on the principle that liquid levels equalise in connected containers.✓ The top and bottom of the gauge glass are connected to the tank with pipes.✓ The liquid moves freely between the tank and glass✓ so the liquid levels are the same in both the tank and the glass.✓ The level-measurement reading on the glass tube✓ has a scale printed on it to indicate the level measurement.✓

OR

It consists of a graduated transparent glass tube✓ connected through valves✓ to the side of the tank of which the level is required.✓ As the liquid level changes inside the tank✓ so does the level in the glass tube.✓ The valves allow for cleaning of the tube and routine maintenance.✓ (6)

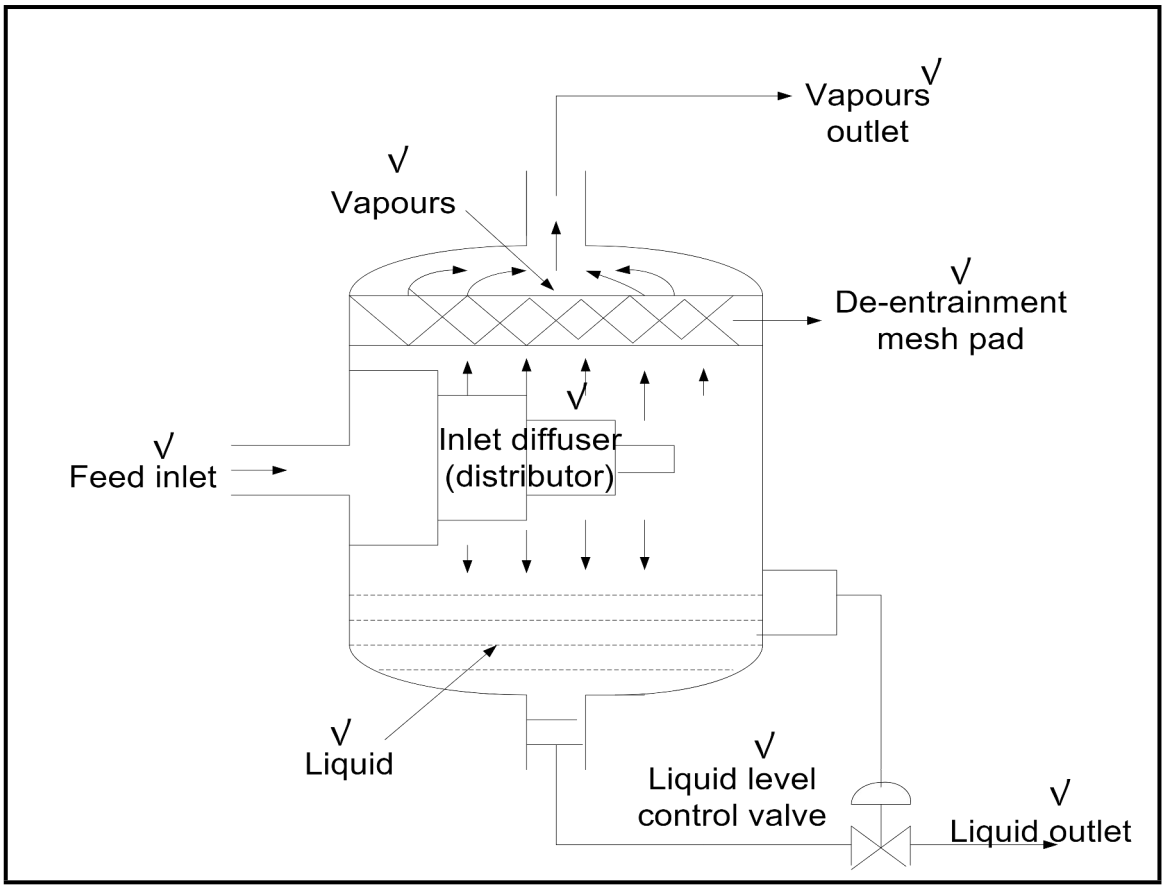
5.2 A solid thermostat is used as the sensory element of a thermal sensor.✓ It consists of two different materials, namely one of the rod and one of the tube of which the expansion rate of the tube material is much higher than that of the rod material.✓ When the temperature rises the tube expands much faster than the rod✓ pulling the rod into the tube thereby activating the switch mechanism that controls the temperature of the water.✓ If the materials are at the same temperature the voltage will be zero.✓ This will increase as the temperature of one metal changes relative to the other until the highest possible value is achieved.✓

OR

The temperature-sensitive portion of this instrument consists of an Invar rod encased in a concentric brass tube.✓ The lower end of the brass tube is spot welded onto the end of the Invar rod. The other ends are left free of each other.✓ When this combination or stem is heated✓ the Invar rod will expand more than the brass case so that the position of the free end of the rod relative to the end of the tube will change.✓ This change in relative position is used to operate a sensitive microswitch.✓ As the stem length increases the differential temperature required gets less.✓ (6)

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5.3



A separator drum is a vertical container into which a liquid and gaseous mixture is fed.✓ Gravity separates the liquid from the gas✓ whereafter the liquid is pushed to the bottom of the container.✓ The gas rises to the top at a specific velocity✓ to make sure that no liquid is trapped in the gas.✓ (13)

5.4 A bimetallic thermometer is made of different metal bars joined together.✓ Because the metal bars are different,✓ and as a result have different coefficients of linear expansion,✓ the top is curled in a helical pattern when they are heated.✓ A pointer is connected to the strip for measurement.✓ When the heat is applied the metals will expand at different rates.✓ The metal bar with a higher expansion rate will expand faster✓ thus causing the strip to curl around.✓ The pointer will be deflected✓ proportional to the amount of heat applied.✓ (10)

5.5
- Checking instruments, i.e. zero checking and calibration
- Repairing or replacing instruments
- Cleaning instruments (3 × 2) (6)

[41]

QUESTION 6

6.1 An electrical motor inside the container rotates the paddle.✓ When the solid level in the container is at a certain height it stops the paddle from turning✓ which signals that the level has reached that point.✓ (3)

6.2
- Define the problem clearly.
- Collect information about relevant changes.
- Develop possible causes.
- Develop solutions.
- Rectify the problem.
- Retest the system. (6)

6.3 **Advantages**
- Simple in operation
- No moving parts
- Long-term reliability
- Inexpensive/Cheap

Disadvantages
- Square root relationships
- Poor turndown rates
- Critical installation requirements
- High irrecoverable pressure loss (4 + 4) (8)

6.4 **Advantages**
- Simple in operation
- Low head loss
- Tolerance of high solids content
- Long-term reliability
- No moving parts

Disadvantages
- Expensive
- Square root pressure velocity relationship
- Poor turndown rates
- Critical installation requirements (4 + 4) (8)

6.5 Control loops control the manufacturing process✓ and influence the quality of products while they are manufactured.✓ This involves collecting data on the process of a product✓ comparing it to the set point using this information to decide if the process needs to be corrected.✓ (4)

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TOTAL: 100

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